ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-1 **WINTER SEMESTER, 2022-2023**

DURATION: 30 MINUTES **FULL MARKS: 15**

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all** **4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. Suppose, R is an arbitrary number. Prove that the ***Newton-Raphson*** formula for finding the square root of R, which is mathematically denoted as $\sqrt{R}$, is – 3 (CO1) (PO2)

$$x_{i+1} = \frac{1}{2}\left(x_i + \frac{R}{x_i}\right)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} & x = \sqrt{R} \\ \Rightarrow\quad & x^2 = R \\ \Rightarrow\quad & x^2 - R = 0 \end{aligned}$$

We take $f(x) = x^2 - R$ and so $f'(x) = 2x$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^2 - R}{2x_i} \\ &= x_i - \left(\frac{x_i}{2} - \frac{R}{2x_i}\right) \\ &= x_i - \frac{x_i}{2} + \frac{R}{2x_i} \\ &= \frac{x_i}{2} + \frac{R}{2x_i} \\ &= \frac{1}{2}\left(x_i + \frac{R}{x_i}\right) \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 2 points for correct derivation.

2. a) What are significant digits? 1 (CO1) (PO1)

Solution:
Significant figures (also known as the significant digits, precision, or resolution) of a number in positional notation are digits in the number that are reliable and necessary to indicate the quantity of something. These are the digits that are known reliably in a reported number.
Rubric:

- 1 point for a correct definition. (Doesn't need to be exactly as written above.)

b) Determine the *minimum* number of significant digits in each of these numerical values — 3 (CO3) (PO2)

(a) 69.0 (b) 69420 (c) 069420
(d) 0.069420 (e) 69.00420 (f) 6.94200×10^5

Solution:
We need to determine the *minimum* number of significant digits.

(i) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.

(ii) 69420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.

(iii) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.

(iv) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.

(v) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.

(vi) 6.94200×10^5 has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.

Rubric:

- 0.5 point for each correct answer. (The reasons are not required.)

3. Compare and contrast the following root-finding numerical algorithms— ***Bisection*** method, ***Newton-Raphson*** method, ***Secant*** method, and ***False Position*** method. 3 (CO3) (PO2, PO4)

Solution:

Any 3 valid comparisons will suffice. Some comparisons can be—

Criteria	**Bisection**	**Newton-Raphson**	**Secant**	**False Position**
Number of initial guesses	2	1	2	2
Condition for initial guesses	Must satisfy $f(x_l)f(x_u) < 0$	Random	Random	Must satisfy $f(x_l)f(x_u) < 0$
Approximation formula	$x_m = \frac{x_l+x_u}{2}$	$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$	$x_{i+1} = x_i - \frac{f(x_i)(x_i-x_{i-1})}{f(x_i)-f(x_{i-1})}$	$x_r = \frac{x_u f(x_l)-x_l f(x_u)}{f(x_l)-f(x_u)}$
Category/Type of method	Bracketed	Open	Open	Bracketed
Guarantee of convergence	Guaranteed	Not guaranteed	Not guaranteed	Guaranteed

Rubric:

- 1 point for each valid comparison.

4. The *Golden Ratio*, often denoted by the Greek letter ϕ (phi), is the ratio of two quantities if their ratio is the same as the ratio of their sum to the larger of the two quantities. Expressed algebraically, for quantities a and b with $a > b > 0$, the ratio $\frac{a+b}{b} = \frac{a}{b} = \phi$ 5 (CO2) (PO3)
The constant ϕ is equal to the positive root of the quadratic equation $x^2 = x + 1$, which is

$$x_{root} = \phi = \frac{1+\sqrt{5}}{2} = 1.618033988749\ldots$$

Use the ***Bisection Method*** to estimate the value of the golden ratio ϕ. The initial upper and lower guesses are $x_l = 1$ and $x_u = 2$ respectively. Conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m), at the end of each iteration. Fill out the following table after performing the necessary calculations.

Iteration	x_l	x_u	x_m	$\|\epsilon_a\|\%$	m	$f(x_m)$
1	1	2		–	–	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_m	$\|\epsilon_a\|\%$	m	$f(x_m)$
1	1	2	1.5	–	–	-0.25
2	1.5	2	1.75	14.286	0	0.3125
3	1.5	1.75	1.625	7.692	0	0.015625
4	1.5	1.625	1.5625	4	1	-0.121

Rubric:

- 0.25 point for each correct answer.